linkedin.com/urisalazar
devpost.com/salazaru
github.com/salazaru

Uri Salazar

(619) 513-6750
uri.salazar@pm.me
urisalazar.com

> displayResume(Education, WorkExperience, Projects, Skills, Status);

// ==========

San Diego, CA **University of California, San Diego** **Sept 2019 – Present**

* **Major**: Electrical Engineering: Machine Learning & Controls, B.S. (September 2022) (Provost Honors) (1st Gen)
* **Courses**: Design Patterns, Parallel Prog., Comp. Archt., Data Analysis, DSP, LNL Optimiz., Pattern Recogn, ASIC

Riverside, CA **Riverside City College** **Feb 2014 – May 2019**

* **Major**: Computer Programming, Cert., C++/Java Programming Cert. (**GPA**: 3.72) (Alpha Gamma Sigma Honors Sociey)
* **Courses**: OOP, Algorithms, Data Structures, Discrete Structures, Operating Systems, Web Development, Game Logic

// ==========

Audio Captioning Programmer **University of California, San Diego** **March 2021 – Present**

* **Project**: Sensing Everyday Activities Using Wi-Fi: A Machine Learning Based Framework - **Mentor**: Dr. Xinyu Zhang
* Utilized PyTorch to train a model that generates egocentric audio captioning based off user-POV GoPro video.
* Wrote a python script to extract over 700GB of audio from the EPIC KITHENS dataset via FFmpeg and Bash.
* Features were extracted from mp3 files to NumPy arrays (.npy) and connected to labels for writing the caption-generator model (.pt); the model was tested against the sample captions from the Clotho v2 dataset.

Supplemental Instructor **Riverside Academic Support** **July 2017 – June 2019**

* Restored underrepresented student graduation rates by composing weekly additional lectures in physics & computer science courses; utilized systems such as Mathematica to visualize and explore problems in detail.
* Restructured the dependency cycle between student and prof. by facilitating independent learning strategies.

Image Processing Researcher **University of California, Riverside** **June 2015 – Sept 2015**

Project: Research Experiences in Integrated Computational Entomology - **Investigator**: Dr. Eamonn Keogh

* Detected Aedes Aegypti mosquitoes in a laboratory environment using computer vision, image segmentation, and histogram modification to aid in the prevention of the dengue virus in rural South America and Africa.
* Incorporated statistical data using R and MATLAB image processing to increase the efficiency of drone-delivered insecticides by predicting precise geographical locations of high mosquito activity/infestations.

// ==========

Prospect: Mobile Application --> https://devpost.com/software/prospect **April 2022 – Present**

* A travel logging journal that recommends destinations based on COVID-19 data for safe tourism & mental health.
* Developed back-end using Spring Boot web framework with PostgreSQL database for storing/processing GPS data.
* Implemented JSON parser for more efficient data collecting; initaiated a GraphQL prototype for map coloring.

Susumeru: Anime Recommender --> https://github.com/salazaru/Susumeru **Nov 2020 – Present**

* Trained a model using a crawled MyAnimeList.net dataset of over 300,000 users and 80 million records.
* Utilized K-nearest neighbors and cosine similarity algorithms to find similarities between users and genres.
* Determined most popular series by country and age; analyzed genre popularity over time through user score.
* Cleaned and filtered distorted data; developed in a remote and collaborative 5-person team environment.

SignVision: Sign Language Translator --> https://devpost.com/software/signvision **January 2020**

* Two-Way translator for sign language assistance using computer vision | H.A.R.D. Hacks 2020 (hackathon)
* Incorporated AlwaysAI computer vision models to translate hand gesture to ASL output via neural network.
* 3D printed a fully-functioning robotic hand with finger movement driven via servo motors and Raspberry Pi3.
* A desired word can be input into the program and the robotic hand will physically demonstrate the sequence.

Positive Edge: Line-tracking Car --> https://github.com/salazaru/plus-edge **Sept 2019 – Dec 2019**

* Microcontroller car based on reinforcement learning | 2nd Place UCSD Robotics Competition (seeding bracket)
* Utilized C++ and various Arduino libraries to input photosensor data into a proportion-integral-derivative (PID) error system that modified dyadic rear-wheel velocities depending on input data (brightness values.)
* Implemented software testing, debugging, and test-driven development around the Arduino Mega 2560 unit.

yourTA Online: IoT Detection System --> https://devpost.com/software/yourta **October 2019**

* "Your Trash Assistant: Online" – San Diego Hacks 2019 (48-hour hackathon) | Category Award Winner
* Developed a sensor-based recycling scoreboard concept for UC campuses by constructing a prototype circuit that included multiple sensors to measure the frequency and volume of proper recycling practices by campus.
* Developed a Node.js server to display updated sensor values in real-time via the ESP8266 Wi-Fi controller.
* Won the Northrop Grumman Award for "Best IoT Device that Incorporates Multiple Nodes."

Syrus: AR Navigation Assistant --> https://github.com/Stents-/SyrusSUITS **Sept 2018 – April 2019**

* **Program**: Spacesuit User-Interface Technologies for Students (SUITS) | NASA Design Challenge Winner
* Developed a navigation system that included pathfinding, virtual arrows, user interface, and a 3D mini-map in C# and Unity to train and assist astronauts aboard the International Space Station complete objectives.
* Programmed using Microsoft HoloLens framework; utilized Leap Motion Controller for optical hand-tracking.

// ==========

Languages: {C++, Python, Solidity} **Familiar**: {C, Java, JavaScript, SQL, HTML, CSS, MATLAB, R, CUDA, C#, Verilog}

Technology: PyTorch, MERN Stack, SaaS, AWS, git, Docker, REST APIs, HTTP Methods, NET Protocols, gtest, Linux

Spoken Languages: English, Spanish. **Beginner**: Japanese（日本語: 初心者） **Citizenship Status**: U.S. Citizen